

SPL1 Project Proposal Form, 2026
Institute of Information Technology (IIT)
University of Dhaka

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Project Description: Title: FaceMatrix Objective. This project proposes to design and implement, from scratch in native C++, a system that detects a human face in an image and recognizes its facial expression. The recognition approach follows the eigenface/PCA methodology of Tawhid, Laskar and Ali, "A Vision-based Facial Expression Recognition and Adaptation System from Video Stream" (IJMLC, 2012). Only the C++ standard library will be used; OpenCV, Eigen, and all other external vision, linear-algebra and machine-learning libraries or pre-trained models are excluded. Planned pipeline. The system will be organized as three stages. (1) <i>Image processing</i>: BMP loading, grayscale conversion, bilinear resizing, contrast stretching, histogram equalization and normalization to a fixed size. (2) <i>Face detection</i>: a multi-scale image pyramid with sliding-window template-similarity matching and non-maximum suppression, giving a face bounding box and crop. (3) <i>Expression recognition</i>: each normalized face is flattened into a vector; the mean face and centered vectors are computed; principal components (eigenfaces) are obtained with a custom eigenvalue solver (e.g., power iteration with deflation or the Jacobi method) and the top K are retained; an input face is projected into the eigenspace and classified by minimum Euclidean distance, with an optional configurable threshold to reject unknown expressions. Dataset. The real FER2013 dataset (BMP format) will be used, preserving its official training/test partitions; a validation set will be drawn from the training partition only. As in the reference paper, five expressions will be recognized: Neutral, Happy, Sad, Surprise and Disgust. Angry and Fear will be explicitly excluded, with excluded counts reported, rather than merged into other classes. No synthetic data will be generated. Evaluation. Performance will be measured on held-out test data using accuracy, the confusion matrix, and per-class precision and recall. Controlled experiments will vary the number of eigenfaces K, the preprocessing steps, image resolution and the rejection threshold, with tuning done on validation data only. Development plan. (1) FER2013 loading and five-class filtering; (2) vectorization and mean-face computation; (3) PCA and eigen-solver with orthogonality checks; (4) eigenface generation, saved as BMP images; (5) eigenspace projection; (6) distance-based classifier and threshold handling; (7) evaluation and experiments; (8) integration with the face detector for end-to-end prediction from an input image; (9) unit tests and documentation. Expected outcome. A modular, tested command-line C++ application that outputs the mean face, eigenfaces, eigenvalues, the predicted expression with distance information, and quantitative evaluation results.			
Languages or Tools to be used: C++ (standard library only; no OpenCV, Eigen or other external CV/ML libraries); GNU Make / CMake for building; custom BMP image I/O, PCA and numerical linear-algebra routines; FER2013 dataset (BMP format) for training and testing; C++ unit tests for each module.			
Supervisor's Name: <u>Dr Md. Nurul Ahad Tawhid</u> Signature of the supervisor: _____ Date: _____			

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback:**Before the Final Presentation:**

I confirm that the project is complete and I am forwarding it for final presentation.

Signature of the supervisor: _____

Date: _____

